Felipe Nunes

+1 (778) 839-8344 — felipenunes4822@gmail.com — linkedin.com/in/felipemeinerznunes — github.com/felipflop

Summary

Integrated Engineer **Education**

University of British Columbia Sep 2020 – May 2026 (Expected)
Bachelor of Applied Science in Integrated Engineering Vancouver, BC **Professional Experience**

Miru Smart Technologies — Systems Integration Intern May 2024 – Aug 2024
Vancouver, BC

- Reduced PCB footprint by 60% (6-layer, BGA SoM) and added a tuned 2.4 GHz inverted-F antenna. :contentReference[oaicite:1]index=1
- Wrote C modules in Zephyr to drive an electrochromic windows controller, increasing transition rate by 50%. :contentReference[oaicite:2]index=2
- Devised a custom compilation pipeline using CMake and linker scripts targeting ARM64. :contentReference[oaicite:3]index=3

Sarcomere Dynamics — Electrical & Firmware Engineering Intern May 2023 – Apr 2024
Vancouver, BC

- Implemented embedded C firmware with a finite-state machine to control actuators in a multi-DOF robotic hand. :contentReference[oaicite:4]index=4
- Redesigned three PCBs with emphasis on USB-C integration, mixed-signal routing, and power delivery. :contentReference[oaicite:5]index=5
- Developed modular C++ classes and extended a custom Python API to support wireless communication (Wi-Fi). :contentReference[oaicite:6]index=6
- Performed reflow soldering, signal probing, and hardware bring-up for embedded prototypes. :contentReference[oaicite:7]index=7

Projects

Robotic Chessboard Sep 2024 – Apr 2025

- Won the Best Engineering Project award across all final-year Capstone projects. :contentReference[oaicite:8]index=8
- Designed custom PCBs and cable assemblies for a multiplexed Hall-effect sensor grid to detect piece positions. :contentReference[oaicite:9]index=9
- Selected stepper motors/drivers; configured and compiled a custom GRBL build for an H-bot gantry. :contentReference[oaicite:10]index=10
- Built hardware abstraction layers; implemented backend polling and state detection algorithms. :contentReference[oaicite:11]index=11

Mountable Heads-Up Display Oct 2022 – Apr 2023

- Engineered an ESP32-based control unit interfacing with UBC Formula Electric's CAN bus. :contentReference[oaicite:12]index=12
- Displayed real-time telemetry (speed, temperature, lap time) on an I2C OLED using FreeRTOS task scheduling. :contentReference[oaicite:13]index=13

- Designed a compact custom PCB integrated into a 3D-printed helmet-mounted enclosure. :contentReference[oaicite:14]index=14

Automated Drink Dispenser Nov 2021 – Apr 2022

- Constructed a custom beverage dispenser using CNC-milled aluminum and a 3D-printed nozzle. :contentReference[oaicite:15]index=15
- Controlled dosing with solenoid valves and load-cell feedback for precision pouring. :contentReference[oaicite:16]index=16
- Wrote firmware for automated calibration, stepper control, drink selection/dispensing, and an LCD UI. :contentReference[oaicite:17]index=17

Engineering Design Teams

Open Robotics — Grippers Team Lead Sep 2024 – Present

- Directed full-cycle development of a parallel gripper end effector for RoboCup@Home; managed an 8-person team from design to integration. :contentReference[oaicite:18]index=18

Supermileage — Electrical Team Member Sep 2022 – Aug 2023

- Redesigned and assembled a custom PCB leveraging analog sensors to detect faults in a hydrogen electric FSAE car; team won the 2023 SEMA hydrogen fuel cell category. :contentReference[oaicite:19]index=19

Technical Skills

Operating Systems: Linux (Arch, Ubuntu, Raspbian), Windows, macOS, FreeRTOS, Zephyr :contentReference[oaicite:20]index=20
Programming: C/C++/C#, Python, JavaScript, HTML/CSS, MATLAB, R, Shell, Make, CMake :contentReference[oaicite:21]index=21
Hardware Design: Circuit design/analysis, PCB layout (Altium, KiCad), FPGAs, VHDL, Multisim, SPICE :contentReference[oaicite:22]index=22
Mechanical: Fusion 360, Onshape, SolidWorks, Orca Slicer :contentReference[oaicite:23]index=23
Tools & Test: Oscilloscopes, Logic Analyzers, Function Generators, Reflow & hand soldering :contentReference[oaicite:24]index=24 **Languages**

French (DELF B2) — Portuguese (Bilingual Proficiency) — Spanish (Working Proficiency) :contentReference[oaicite:25]index=25